Project Progress Report

- ANGELES MATHENA (A0351915A)
- ALZAMIL NOJOOD AHMAD (A0350755B)
- SU YUBO (A0356908N)
- SZENTPETERI DAVID (A0357858E)
- DAI PEISON (A0351174J)

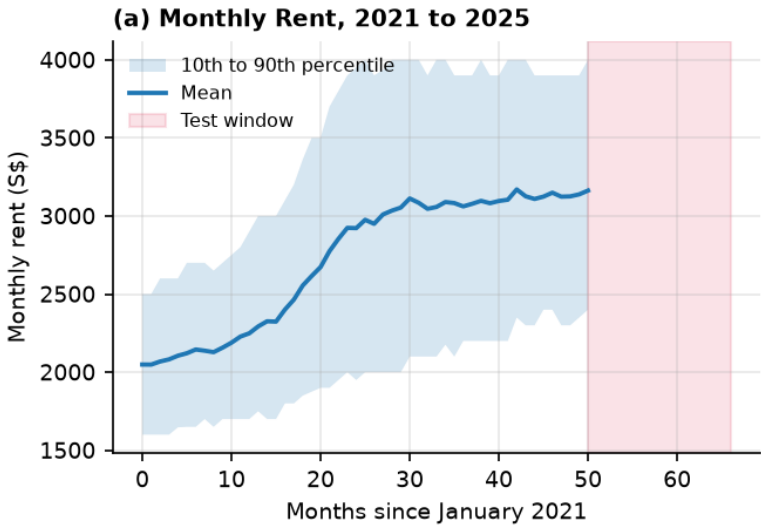
September 30, 2026

Project Overview

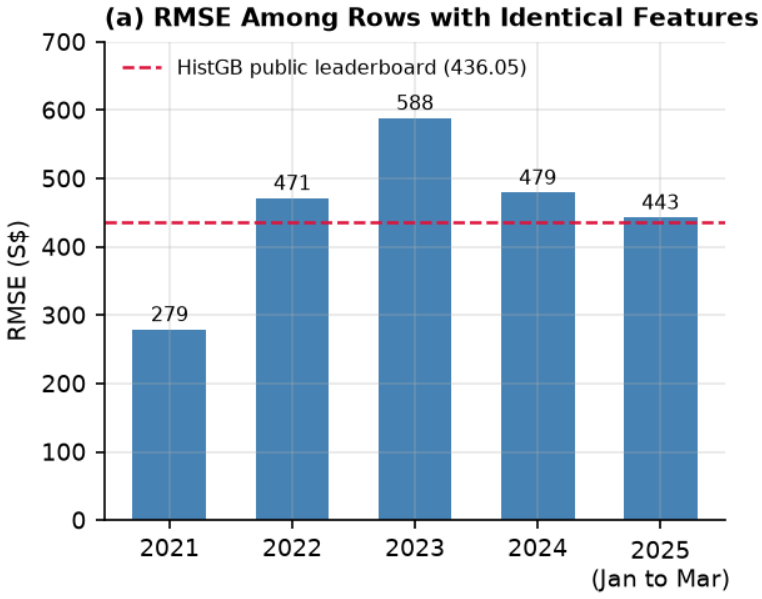
- **Project Motivation:** Renters want to know what their money buys, landlords want to price flats well, and the government tracks affordability. HDB rents rose 54% over 2021–25. A useful model must track where the market is going.
- **Problem Statement:** Predict `MONTHLY_RENT` for 50,000 rentals from 2025-03 to 2026-07, given 150,000 labeled rentals from 2021-01 to 2025-03. Predictions are scored by RMSE. The test window starts in the last labeled month and runs 16 months past it, so the market level must be forecast.
- **Project Objectives:**
  1. How should the market level be forecast up to 16 months ahead?
  2. Does encoding location from past rents help once coordinates are known?
  3. Which feature groups carry signal, and does the model family matter?
  4. How do we validate so that the score reflects the real horizon?
  5. What drives rent once the market level is removed?

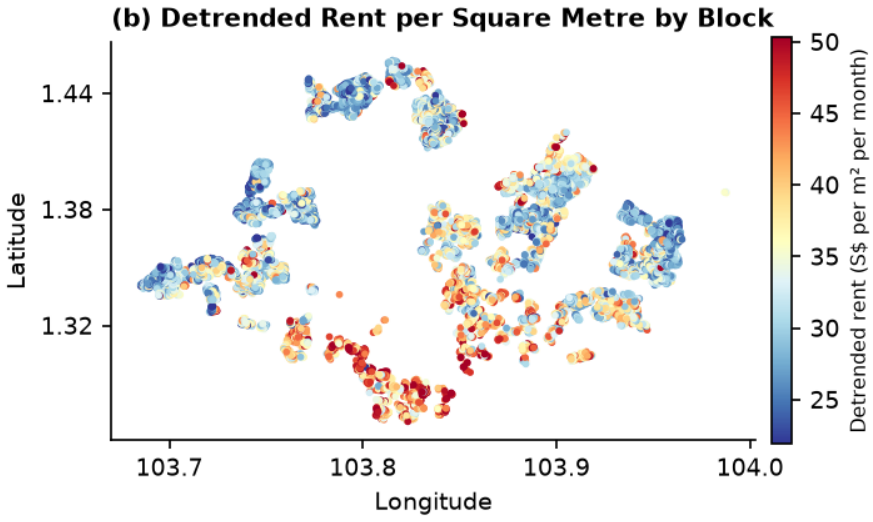
Dataset

- **Dataset Description:** Each row is one rental approval, with its town, street, block, flat type, flat model, floor area and lease start. Auxiliary files give block coordinates, MRT stations, malls, schools, COE and stock prices.
- **Number of Records:** There are 150,000 train and 50,000 test rows across 26 towns and 576 streets.
- **Important Features:** Month of year and location matter most, as the Results slides show. Mean rent rose from S$2,049 to S$3,163 (**+54%**) and then plateaued, which shapes the whole design.



Dataset: How Much Is Predictable?





- Rows with the same block, flat type and month still differ. The RMSE among them is 279-588 depending on the year, and 443 in 2025 (January to March 2025). This is a rough noise estimate, not a strict floor, and our 431.01 is already below it.
- Detrended rent per square meter shows a clear central premium, so location has to be modeled carefully.

Methodology

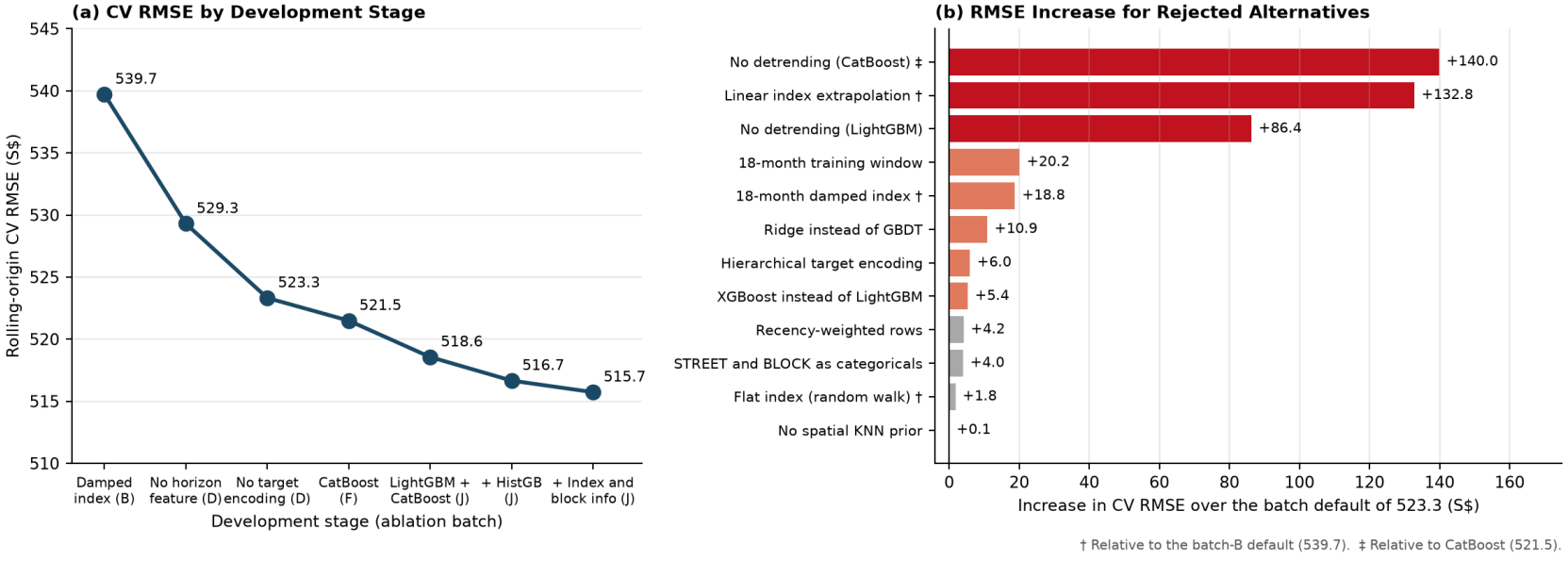
Rent is factorized into a market level and a flat-specific part, $\text{RENT}_{it} \approx \text{BASE}_i \times \text{INDEX}_t$. A tree cannot predict outside the range of its training target, so without the index it keeps predicting old prices.

- **Data Preprocessing:** `FEE` and `FURNISHED` are constant, so we drop them, and we normalize text fields and flat-type spellings. The only missing values come from 159 of 200,000 rows that are unmatched in the HDB Property Information register, and we leave them as NaN for the trees.
- **Feature Engineering:** For open and planned MRT stations, malls and schools, we compute the distance to the nearest three and the counts within 0.5, 1 and 2 km. We also add the distance to the CBD, lease age, area per room, block height and block composition (units, flat mix and share of HDB public-rental-scheme units). The register has no rents and no dates.
- **Index:** The index is fitted by alternating least squares and forecast with a damped trend (12-month window, damping 0.6). We fit one index per flat type and shrink it halfway toward the global one.
- **Models Tested:** We test LightGBM, CatBoost, XGBoost and Ridge on the detrended target, and HistGB on raw rent with its own features and block rent history. The submission blends $0.7 \times (0.5\,\text{LightGBM} + 0.5\,\text{CatBoost}) + 0.3 \times \text{HistGB}$.

Methodology: Evaluation Protocol

- **Rolling Origin:** We train on the months before a cut at month 27, 31 or 35, then score the next 16 months, as in the test. The index is refitted in every fold, so each fold pays for a wrong price level. We do this because the test split lies later in time, not because the data is a time series.
- **Cut-39 Check:** We also refit at month 39 and score months 39–50. A candidate is kept only if it also improves there. These months lie inside fold 35's window and the check was used to accept or reject candidates.
- **Ablations:** Each changes one component against a fixed reference, as a named configuration that is run locally or as a SLURM array.
- **Integrity:** The public 2021 rental file that the competition was cut from is never joined, because it contains the test labels.

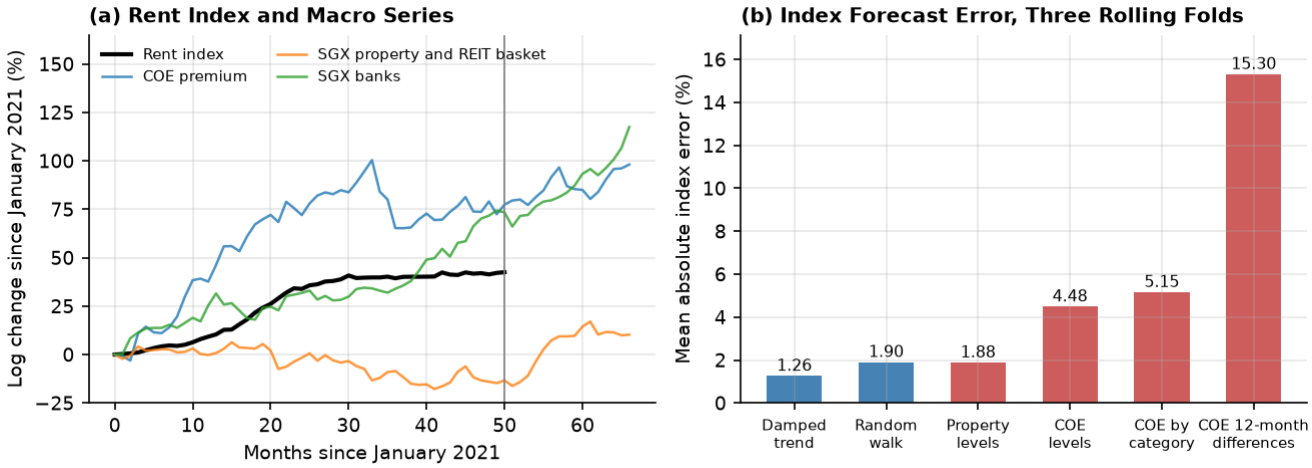
Results: Where the Gains Came From



- Removing the index costs more than any other choice. We got +86.4 RMSE for LightGBM and +140.0 for CatBoost, because the trees keep predicting old prices.
- Model class matters much less. With the trend removed, Ridge is only 11 behind LightGBM.
- Each point in (a) is the best setting of its batch, and each bar in (b) is measured against its own batch's default.

Results: Forecasting the Market Level

Index Forecast (LightGBM)	CV RMSE	Bias
Damped, 12 m, 0.6	**539.7**	+6.7
Flat	541.5	−46.4
Damped, 18 m, 0.85	558.5	+127.4
Linear, 18 m	672.6	+340.8



- A trend that persists too long over-predicts badly. A short, strongly damped trend wins, but only by 1.8 RMSE over a flat index. Its real gain is bias, which falls from −46.4 to +6.7.
- COE premiums correlate 0.91 with the index in levels, yet forecast it worse than a random walk. Two rising series correlate in levels without either predicting the other.

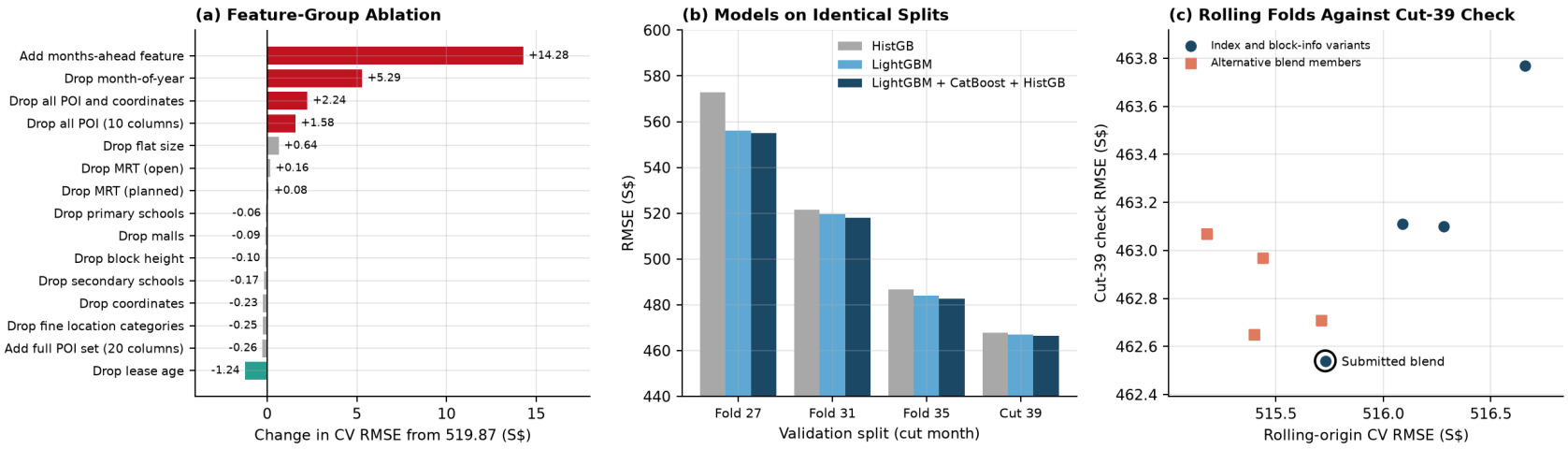
Results: Target Encoding Made Things Worse

Hierarchical target encoding replaces each block with its past detrended rent, shrunk toward its street and town.

Configuration	CV RMSE	Bias
With target encoding	529.3	−18.0
Without it	**523.3**	−1.5
Without the spatial prior too	523.4	−0.5
STREET, BLOCK as categoricals	527.3	+0.1

- Dropping the encoding gains 6 RMSE and removes most of the bias.
- **Is it staleness?** A lagged encoding, as out of date in training as at test time, is 12.5–15.2 worse than no encoding at all.
- Our reading is that coordinates and POI features already capture location, so the encoding adds noise.

Results: Features, Head-to-Head, and Selection



- **(a)** POI groups substitute for each other. Any one group alone moves RMSE by at most 0.17, while all ten together move it by 1.58. Dropping lease age is the best of 15 changes (-1.24), but we keep lease age until the drop is confirmed on CatBoost.
- **(b)** HistGB loses to LightGBM on every split but helps in the blend. On the public leaderboard, the blend scores **431.01** against 436.05 for HistGB alone.
- **(c)** Several alternative blends beat the shipped one on the rolling folds but not at cut 39, so we kept the simpler blend.

Challenges

- **Small gains do not survive a second check.** Six alternative blends (with other weights, a Ridge member, or a HistGB member given our features) beat the shipped blend on the folds by up to 0.55. All six were worse at cut 39, so we kept the simpler blend.
- **Retuning buys little.** The best LightGBM retune gained 0.24 RMSE and took 4.4 times as long to fit.
- **The level forecast is the unstable part.** The LightGBM + CatBoost blend's bias swings between folds, from -37.3 to $+48.8$ to -27.7, and it is -42.2 at cut 39.

Next Steps

- **Model Refinement:** We will select on folds 27 and 31 only and keep months 47–50, which no selection step has seen, as a clean final check that we score once. On that setup we will test a horizon-aware index correction and confirm or reject the lease-age drop.
- **Further Evaluation:** We will break residuals down by months ahead, flat type, town and unseen blocks, and study feature importance and partial dependence on the detrended target.
- **Final Report and Presentation:** The report will cover methods, evaluation and limitations, with related work on damped trends (Gardner & McKenzie, 1985) and rolling-origin evaluation (Tashman, 2000).